

📦 Offline Debian/Ubuntu Package Cache Guide 📦

☆☆ Introduction ☆☆

This guide provides comprehensive instructions on how to prepare a full package cache for Debian/Ubuntu systems, enabling offline installation of .deb packages. This is particularly useful for environments with limited or no internet connectivity, or for maintaining consistent software versions across multiple machines. We will explore several methods, from simple apt commands to dedicated mirroring tools.

1. 📁 Method 1: Using apt --download-only 📁

This method is straightforward for downloading specific packages and their dependencies. The downloaded .deb files are stored in /var/cache/apt/archives/.

📋 Steps:

1. **Download Packages:** On your online machine, use the apt --download-only command to download the desired packages.

```
sudo apt update
sudo apt install --download-only <package_name>
# To download a package and its dependencies without installing:
sudo apt-get install --download-only <package_name>
# To re-download already installed packages:
sudo apt --reinstall --download-only <package_name>
```

You can specify multiple package names.

2. **Locate Downloaded Packages:** The .deb files will be in /var/cache/apt/archives/.

```
ls -lh /var/cache/apt/archives/
```

3. **Transfer Packages:** Copy the contents of /var/cache/apt/archives/ to a portable storage device (USB drive, external HDD) and transfer them to your offline machine.

✓ Pros & ✗ Cons:

- **Pros:** Simple, no extra tools needed, good for a few specific packages.
 - **Cons:** Not suitable for mirroring an entire repository, requires manual dependency management if not using apt install --download-only.
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2. 🌀 Method 2: Using apt-mirror 🌀

apt-mirror is a powerful tool for creating a local mirror of entire Debian/Ubuntu repositories.

📋 Steps:

1. **Install apt-mirror:** On your online machine, install apt-mirror.

```
sudo apt update
sudo apt install apt-mirror
```

2. **Configure apt-mirror:** Edit the configuration file `/etc/apt/mirror.list`.

```
sudo nano /etc/apt/mirror.list
```

Uncomment and adjust the set `base_path` to your desired mirror location (e.g., `/mnt/apt-mirror`). Ensure this directory has enough space.

```
set base_path /var/spool/apt-mirror
# set base_path /mnt/apt-mirror # Example for a larger drive
```

Define the repositories you want to mirror. For example, to mirror Ubuntu Jammy Jellyfish:

```
deb http://archive.ubuntu.com/ubuntu jammy main restricted
universe multiverse
deb http://archive.ubuntu.com/ubuntu jammy-security main
restricted universe multiverse
deb http://archive.ubuntu.com/ubuntu jammy-updates main
restricted universe multiverse
# deb-src http://archive.ubuntu.com/ubuntu jammy main restricted
universe multiverse
# deb http://archive.ubuntu.com/ubuntu jammy-backports main
restricted universe multiverse
# deb http://archive.ubuntu.com/ubuntu jammy-proposed main
restricted universe multiverse
```

You can also add lines for Debian or other distributions.

3. **Create Mirror Directory:** Create the base path directory if it doesn't exist.

```
sudo mkdir -p /var/spool/apt-mirror
sudo chown apt-mirror:apt-mirror /var/spool/apt-mirror # Adjust
ownership if needed
```

4. **Start Mirroring:** Run `apt-mirror` to start downloading the repository. This can take a very long time and consume a lot of disk space.

```
sudo apt-mirror
```

You can monitor the progress.

5. **Transfer Mirror:** Once mirroring is complete, transfer the entire mirrored directory (e.g., `/var/spool/apt-mirror/mirror/archive.ubuntu.com/ubuntu/`) to your offline machine.

✓ Pros & ✗ Cons:

- **Pros:** Creates a full, self-contained repository, easy to update incrementally.
- **Cons:** Requires significant disk space and bandwidth, initial sync can be very long.

3. ✗ Method 3: Using debmirror ✗

`debmirror` is another robust tool for creating partial or full mirrors of Debian-based repositories. It offers more fine-grained control than `apt-mirror`.

□ Steps:

1. **Install debmirror:** On your online machine, install `debmirror`.

```
sudo apt update
sudo apt install debmirror
```

2. **Create Mirror Directory:** Create a directory for your mirror.

```
sudo mkdir -p /mnt/debmirror
```

3. **Start Mirroring:** Run debmirror with appropriate options.

```
sudo debmirror \  
  --arch=amd64 \  
  --host=archive.ubuntu.com \  
  --root=/ubuntu \  
  --dist=jammy,jammy-security,jammy-updates \  
  --section=main,restricted,universe,multiverse \  
  --progress \  
  --method=http \  
  /mnt/debmirror
```

- --arch: Specify architectures (e.g., amd64, i386).
- --host: The repository host.
- --root: The path on the host.
- --dist: Distributions/suites to mirror (e.g., jammy, focal).
- --section: Components to mirror.
- --progress: Show progress.
- --method: Protocol to use.
- The last argument is your local mirror path.

This command will download the specified parts of the Ubuntu Jammy repository to /mnt/debmirror.

4. **Transfer Mirror:** Transfer the entire mirrored directory to your offline machine.

✓ Pros & ✗ Cons:

- **Pros:** Highly configurable, can mirror specific parts of a repository, good for custom mirrors.
 - **Cons:** Can be complex to configure initially, still requires significant disk space and bandwidth.
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4. ✈ Method 4: Using apt-offline ✈

apt-offline is designed specifically for managing packages on offline machines by leveraging an online machine to download necessary files.

□ Steps:

On the Offline Machine:

1. **Install apt-offline:** If possible, install apt-offline from a local .deb file or a minimal installation.

```
sudo apt install apt-offline
```

2. **Generate Signature File:** Generate a signature file (.sig) that contains information about the packages needed and the current state of your system.

```
sudo apt-offline set /tmp/apt-offline.sig  
# To include updates for specific packages or the whole system:  
# sudo apt-offline set /tmp/apt-offline.sig --update  
# To include specific packages:  
# sudo apt-offline set /tmp/apt-offline.sig --install-packages  
<package1>,<package2>
```

Transfer /tmp/apt-offline.sig to your online machine.

On the Online Machine:

1. **Install apt-offline:** `bash sudo apt install apt-offline`
2. **Download Packages:** Use the signature file from the offline machine to download all required packages and their dependencies.


```
sudo apt-offline get -d /tmp/apt-offline-downloads /path/to/apt-offline.sig
```

This will download all .deb files into /tmp/apt-offline-downloads.
3. **Transfer Packages:** Copy the entire /tmp/apt-offline-downloads directory to your offline machine.

Back on the Offline Machine:

1. **Install Packages:** Apply the downloaded packages to your offline system.


```
sudo apt-offline install /path/to/apt-offline-downloads
```

This command will automatically install the downloaded .deb files and update apt's cache.

✓ Pros & ✗ Cons:

- **Pros:** Efficient, downloads only what's needed, ideal for truly offline scenarios.
 - **Cons:** Requires apt-offline on both machines, a bit more involved workflow.
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⚙️ Configuring the Offline Machine ⚙️

Once you have your .deb packages or a full mirror, you need to configure your offline machine to use them.

Option A: Using dpkg -i (for individual packages)

If you only have a few .deb files (e.g., from `apt --download-only`), you can install them manually.

```
sudo dpkg -i /path/to/downloaded_packages/*.deb
# If there are dependency issues, try:
sudo apt --fix-broken install
```

You might need to install dependencies in the correct order.

Option B: Configuring apt to use a Local Repository (for mirrored repositories)

If you've used apt-mirror or debmirror to create a full local repository, you can configure apt to use it.

1. **Create a Local Repository:** Assuming your mirrored repository is at /mnt/apt-mirror/mirror/archive.ubuntu.com/ubuntu/.

```
sudo apt-get install apt-ftparchive # If not already installed
cd /mnt/apt-mirror/mirror/archive.ubuntu.com/ubuntu/dists/jammy/
sudo apt-ftparchive packages . > Packages
sudo gzip -c Packages > Packages.gz
sudo apt-ftparchive release . > Release
```

Repeat for jammy-security, jammy-updates, etc.

2. **Add Local Repository to sources.list:** Edit /etc/apt/sources.list on your offline machine.

```
sudo nano /etc/apt/sources.list
```

Add a line pointing to your local mirror. **Comment out or remove any internet-based repositories.**

```
# Example for apt-mirror:
deb file:///mnt/apt-mirror/mirror/archive.ubuntu.com/ubuntu/
jammy main restricted universe multiverse
deb file:///mnt/apt-mirror/mirror/archive.ubuntu.com/ubuntu/
jammy-security main restricted universe multiverse
deb file:///mnt/apt-mirror/mirror/archive.ubuntu.com/ubuntu/
jammy-updates main restricted universe multiverse

# Example for debmirror (adjust path as per your mirror):
# deb file:///mnt/debmirror jammy main restricted universe
# multiverse
# deb file:///mnt/debmirror jammy-security main restricted
# universe multiverse
# deb file:///mnt/debmirror jammy-updates main restricted
# universe multiverse
```

Ensure the path after `file://` points to the root of your mirrored repository (where `dist`s and `pool` directories are located).

3. **Update apt Cache:** `bash sudo apt update` Now, apt will use your local repository for installations.

□ Conclusion □

Preparing an offline package cache is a robust solution for managing software in disconnected environments. Whether you need a few specific packages or a full repository mirror, the tools and methods outlined in this guide provide flexible options to suit your needs. Remember to plan for sufficient storage and bandwidth, especially for full repository mirrors.
